Self-Learning: Red-Black Trees Data Structures

> **Definition:** A **Red-Black Tree** is a self-balancing Binary Search Tree where each node stores an extra bit representing **color (Red or Black)**, satisfying specific balance constraints that ensure the tree height remains bounded by **2 \log_2(N + 1)**.

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1. Detailed Technical Explanation

The 5 Red-Black Tree Properties:

1. **Node Color:** Every node is either **RED** or **BLACK**.

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## 2. AVL Tree vs Red-Black Tree Comparison

| Feature | AVL Tree | Red-Black Tree |

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## 3. Insertion in Red-Black Tree

When inserting a new key:

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## 4. Quick Recall Flow
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